

# Railway Flood-Risk Digital Twin for the SNCF Tartaiguille Corridor

## Validation, Architecture & Scientific Calibration Report

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**Corridor:** Ligne 400 (Montélimar–Marseille), Tartaiguille Section, Drôme, France

**Coordinates:** 44.6559°N, 4.9172°E (Lambert 93: 851987 E, 6397047 N)

## Abstract

This report presents the design, implementation, and validation of a 4-layer railway flood-risk Digital Twin for the SNCF Tartaiguille corridor (Ligne 400, Drôme). The system integrates real-time meteorological data (Open-Meteo API), a hydrological screening layer based on the Soil Water Index (SWI), 2D hydraulic modeling (HEC-RAS), and vulnerability assessment using log-normal fragility curves calibrated against field data from Tsubaki et al. (2016).

The architecture implements a "Funnel Strategy" where the lightweight SWI layer screens the corridor in sub-second time, triggering the computationally expensive HEC-RAS 2D engine only when soil saturation exceeds a critical threshold ( $SWI > 100$  mm). The F5 Grouped Architecture partitions the corridor into 21 sections, consolidating 107 physical assets (track segments, embankments, culverts, and bridges) to evaluate safety thresholds. The system resolves raw grid-level WSE outputs from pre-computed HEC-RAS HDF5 files (Plan 2: 21092025 Cévenol Storm) to generate RAMS-compliant alert levels (GREEN/YELLOW/ORANGE/RED) using a worst-case group roll-up.

A sensitivity analysis of the SWI half-life parameter ( $T = 3\text{--}60$  days) demonstrates model robustness, and a historical Cévenol storm replay validates end-to-end pipeline functionality. This report details the final twin architecture, documents database inventory schemas, addresses map-rendering heuristics (depth-based alpha masking), and outlines a maturity roadmap for future live data assimilation.

**Keywords:** Digital Twin, Railway Infrastructure, Flood Risk, HEC-RAS, SWI, Fragility Curves, F5 Grouping, SNCF, RAMS

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## 1. Introduction

### 1.1 Problem Statement

Railway infrastructure in southern France is increasingly exposed to flash-flood hazards driven by intense Cévenol storm events. The Tartaiguille corridor (Ligne 400, PK 610–660) crosses the Drôme valley, where Mediterranean precipitation can generate rapid surface runoff threatening track stability. SNCF's RISK-VIP program (Cheetham et al., 2016) established that conventional periodic inspection cannot capture the temporal dynamics of flood exposure, necessitating real-time monitoring solutions.

### 1.2 Objectives

This thesis develops a Digital Twin prototype that:

1. **Screens** the corridor in real-time using a Soil Water Index (SWI) to identify saturation conditions.
2. **Simulates** flood hydraulics via HEC-RAS 2D when saturation thresholds are exceeded.
3. **Groups** assets functionally to align hydraulic water surface elevations with structural dependencies (F5 Grouping).
4. **Evaluates** structural vulnerability using fragility curves calibrated against published field data.
5. **Dispatches** RAMS-compliant alerts (speed restriction/halt) to operations control.

### 1.3 Scope

The system is designed for the 50 km Tartaiguille section with:

- **DTM resolution:** 1m (LiDAR-derived, Terrain.lhd\_fx\_lasd1.tif )
- **Temporal resolution:** 15-minute operational cycle
- **Input:** Open-Meteo API (historical + 48h forecast)
- **Output:** RAMS traffic-light alerts per BIM asset and section group
- **Database:** 107 assets mapped across 21 corridor sections

## 2. Literature Review

### 2.1 Digital Twins in Infrastructure

The concept of a Digital Twin (DT) for infrastructure management has evolved from static Building Information Models (BIM) toward dynamic, data-driven systems. Kaewunruen et al. (2021) demonstrated a BIM-embedded digital twin for the Taipei MRT, integrating lifecycle assessment with 3D visualization, but their system lacked real-time sensor fusion. Kim et al. (2025) advanced the state-of-the-art with a stormwater digital twin for Austin's Waller Creek watershed, achieving KGE improvement from 0.633 to 0.786 and sensor fault detection through Extended Kalman Filtering (EKF).

Pedersen (2023) defines three maturity levels:

- **Digital Model:** offline simulation, no live data.
- **Digital Shadow:** unidirectional data flow (physical → digital).
- **Digital Twin:** bidirectional data flow with feedback loop.

Table 1: Comparison of Digital Twin implementations in infrastructure management.

| Reference                | Domain        | DT Maturity    | Data Assimilation | Validation             |
|--------------------------|---------------|----------------|-------------------|------------------------|
| Kaewunruen et al. (2021) | Railway MRT   | Digital Model  | None              | BIM quantities         |
| Kim et al. (2025)        | Stormwater    | Digital Twin   | EKF               | KGE=0.786, AUC=0.99    |
| Cheetham et al. (2016)   | Railway flood | Digital Shadow | None              | 87.5% incident capture |
| This work                | Railway flood | Digital Shadow | None (future)     | See Section 5          |

### 2.2 Soil Water Index (SWI) for Flood Screening

The Soil Water Index quantifies antecedent soil moisture as a precursor to surface runoff. Siva Subramanian et al. (2025) implemented SWI-based thresholds for Japan's Te-LEWS landslide early warning system. The SWI Leaky Bucket model used in this work follows the recursive formulation:

$$SWI(t) = R(t) + SWI(t - 1) \times C$$

Where  $R(t)$  is rainfall intensity (mm/h),  $C = 2^{-1/(T \times 24)}$  is the hourly decay constant, and  $T$  is the half-life parameter (days).

## 2.3 HEC-RAS 2D for Railway Flood Modeling

The Hydrologic Engineering Center's River Analysis System (HEC-RAS) solves the 2D shallow water equations on unstructured grids. HEC-RAS is preferred for overland flood modeling due to its subgrid bathymetry, terrain-following mesh capabilities, and integration with LiDAR-derived DTMs.

## 2.4 Fragility Curves for Railway Vulnerability

Tsubaki et al. (2016) developed the first fragility curves specifically for railway embankment and ballast scour, using data from two Japanese flood events:

- **Ballast scour:** median  $\Delta h = 0.30$  m,  $\sigma = 0.035$  (normal),  $n = 15$  observations.
- **Combined (ballast + embankment):** median  $\Delta h = 0.22$  m,  $\sigma = 0.15$  (log-normal),  $n = 31$  observations.

These curves were validated against aerial photograph flood extents and are adapted here to define structural vulnerability.

# 3. Study Area & Data

## 3.1 Tartaiguille Corridor

The study area covers the Tartaiguille section of SNCF Ligne 400 (Montélimar–Marseille) in the Drôme department (44.65°N, 4.91°E). The corridor traverses a valley susceptible to flash flooding during Cévenol storm events, which can deliver 200+ mm of precipitation within 24 hours.

## 3.2 Digital Terrain Model

Table 2: DTM characteristics.

| Parameter                     | Value                                      |
|-------------------------------|--------------------------------------------|
| File                          | Terrain.lhd_fx_lasd1.tif                   |
| Resolution                    | ~1 m (LiDAR)                               |
| CRS                           | EPSG:2154 (Lambert 93)                     |
| Elevation range               | 167.7 m – 376.3 m                          |
| Track elevation (Voie_seg_00) | 207.11 m (terrain), 207.11 m (ballast top) |

## 3.3 BIM Asset Registry & F5 Sections

The digital twin incorporates 107 BIM assets across 4 categories, segmented horizontally into 21 corridor sections (each spanning ~100m of track).

## Asset Distribution

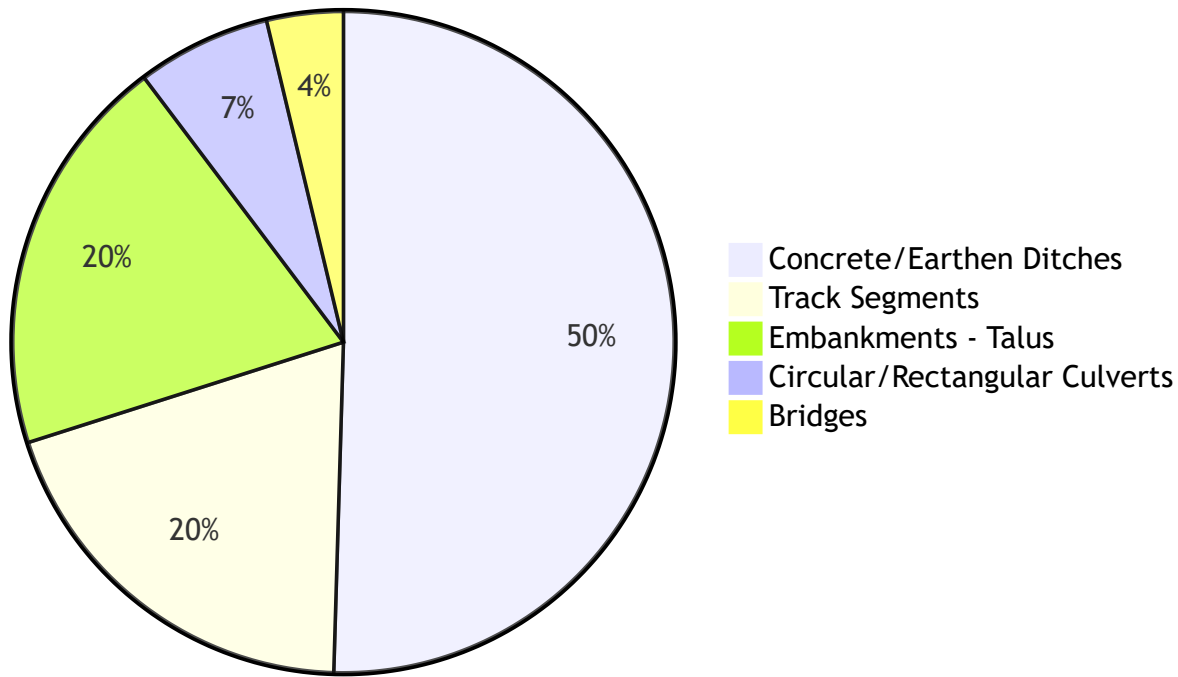


Table 3: BIM asset registry summary.

| Asset Category      | Count | Primary Data Files                                                                          |
|---------------------|-------|---------------------------------------------------------------------------------------------|
| Track Segments      | 21    | voie_segments.json , voie_fixed.gpkg                                                        |
| Embankments (Talus) | 21    | Talus Terre_fixed.gpkg                                                                      |
| Drainage Assets     | 61    | Buse_fixed.gpkg , Dalot_fixed.gpkg , Fossé terre_fixed.gpkg , Fossé terre revêtu_fixed.gpkg |
| Bridges             | 4     | Pont Rail_fixed.gpkg                                                                        |

## 3.4 Asset Elevation Distribution

The elevation distribution of all 107 BIM assets was extracted from the 1m LiDAR DTM using zonal statistics. Figure 1 below shows the box-and-whisker distributions for each asset layer, confirming that the vertical alignment of the asset database is consistent with the underlying terrain model.

Figure 01: Elevation Distribution across Railway Assets (Case Study)

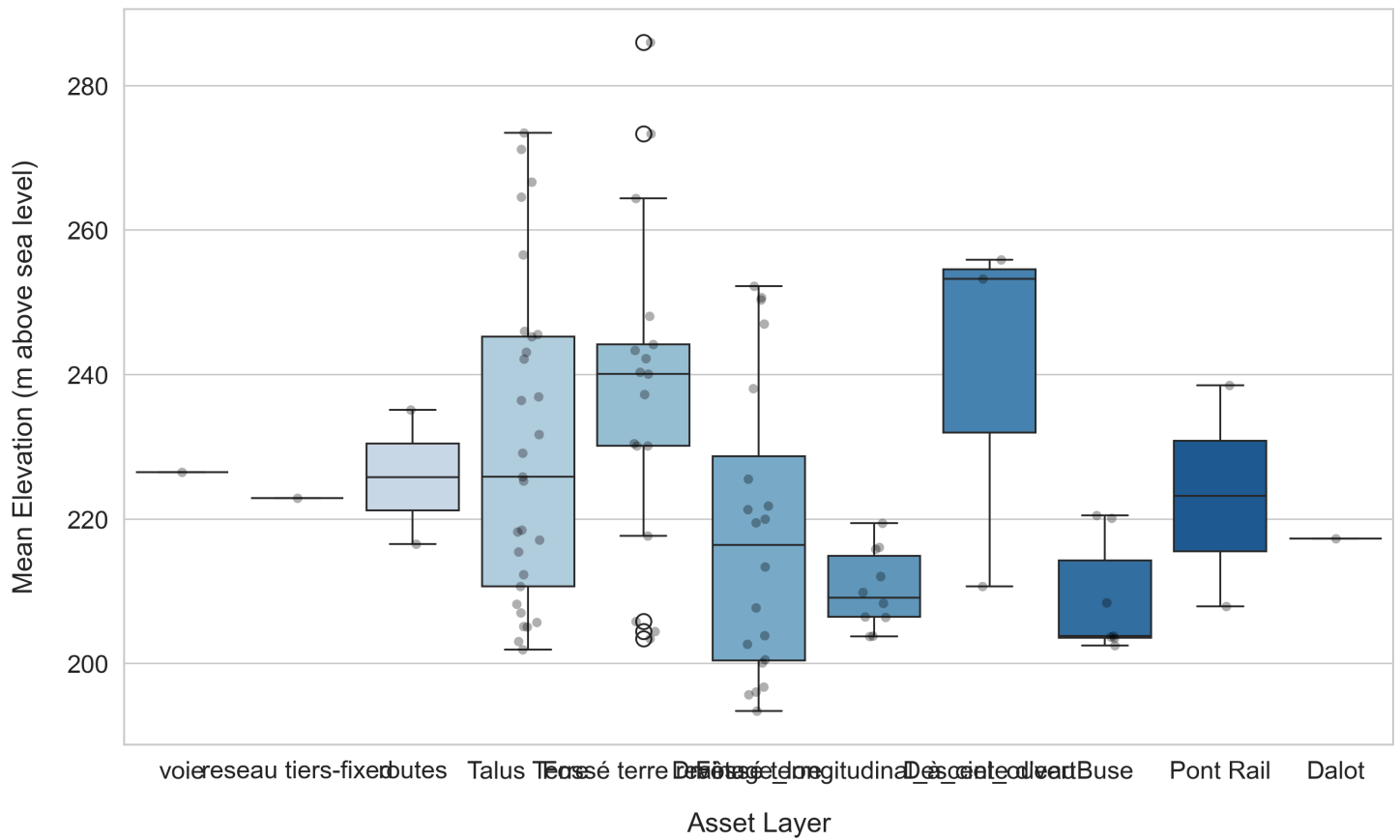


Table 4: Representative asset elevation statistics extracted from the DTM (selected assets shown).

| Asset Layer        | Asset Label          | Elev. Min (m) | Elev. Max (m) | Elev. Mean (m) | Pixel Count |
|--------------------|----------------------|---------------|---------------|----------------|-------------|
| Voie               | Voie_0               | 204.01        | 288.66        | 226.46         | 16,453      |
| Talus Terre        | Talus Terre_0        | 205.61        | 217.02        | 212.28         | 1,822       |
| Talus Terre        | Talus Terre_12       | 203.61        | 214.82        | 208.21         | 6,316       |
| Fossé terre        | Fossé terre_10       | 206.41        | 208.41        | 207.70         | 156         |
| Fossé terre revêtu | Fossé terre revêtu_0 | 238.95        | 244.45        | 240.34         | 75          |
| Buse               | Buse_0               | 203.21        | 203.61        | 203.46         | 30          |
| Buse               | Buse_5               | 203.71        | 203.81        | 203.78         | 35          |
| Pont Rail          | Pont Rail_3          | 204.71        | 210.51        | 207.88         | 194         |
| Dalot              | Dalot_0              | 213.32        | 219.62        | 217.26         | 165         |

### 3.5 3D BIM Asset Inventory

Table 5: Complete 3D BIM MULTIPATCH asset inventory extracted from `data/raw/maquette_3d/`.

| Layer                              | Feature Count     | Z Range (m NGF) | Source        |
|------------------------------------|-------------------|-----------------|---------------|
| Buse (Culverts)                    | 7                 | 201 – 221       | 3D MULTIPATCH |
| Dalot (Box Culverts)               | 1                 | 213 – 216       | 3D MULTIPATCH |
| Fossé terre (Earth Ditches)        | 31                | 175 – 258       | 3D MULTIPATCH |
| Fossé terre revêtu (Lined Ditches) | 22                | 178 – 288       | 3D MULTIPATCH |
| Talus Terre (Embankments)          | 36                | 178 – 284       | 3D MULTIPATCH |
| Voie (Track)                       | 1 (→ 21 segments) | 180 – 290       | 3D MULTIPATCH |
| Descente d'eau (Downpipes)         | 3                 | 210 – 267       | 3D MULTIPATCH |
| Drainage longitudinal              | 10                | 204 – 222       | 3D MULTIPATCH |

### 3.6 HEC-RAS 2D Model Specifications

Table 6: HEC-RAS 2D model specifications and pre-computed simulation plan parameters.

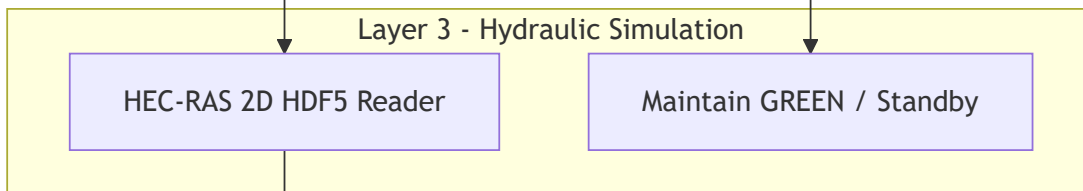
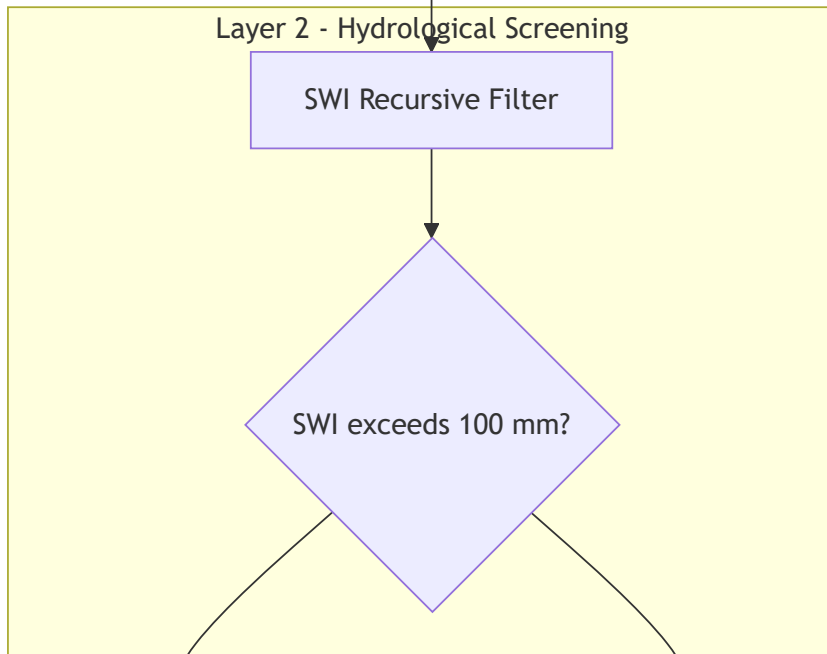
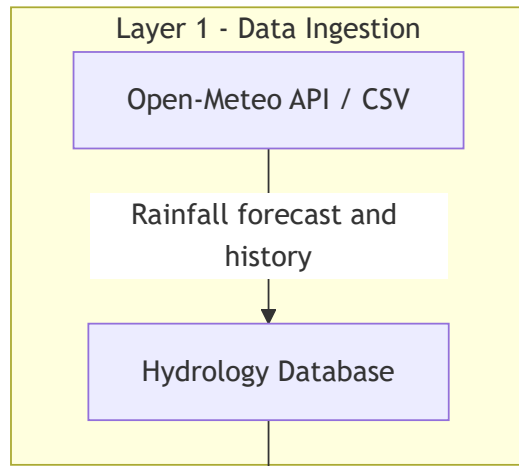
| Parameter          | Value                                                                                  |
|--------------------|----------------------------------------------------------------------------------------|
| Project            | CAPSTONE_JN_L752_PK (Ligne 752, PK534 — South Head Tartaiguille)                       |
| Software Version   | HEC-RAS 6.60, SI Units                                                                 |
| CRS                | EPSG:2154 (Lambert 93)                                                                 |
| 2D Flow Area       | PK534_FA_5M2                                                                           |
| Total Mesh Cells   | 950,122 (~5 m² resolution)                                                             |
| Boundary Condition | PK534_BL , friction slope = 0.01                                                       |
| Structures         | 9 SA/2D connections (culverts)                                                         |
| Plan 1 (p01)       | R100_1HR — 30 MAR 2026 13:00→14:00, 13 timesteps, Δt=1s compute / 5min output          |
| Plan 2 (p02)       | 21092025 — 21 SEP 2025 07:00→22 SEP 04:00, 127 timesteps, Δt=5s compute / 10min output |

## **4. Methodology**

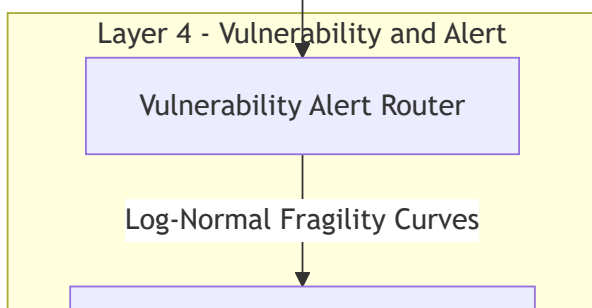
### **4.1 Architecture Overview (4-Layer Funnel Strategy)**

The system architecture implements a "Funnel Strategy" that progressively narrows computational focus. This ensures that the computationally heavy HEC-RAS engine is only loaded when there is a physical threat of runoff.





WSE per asset



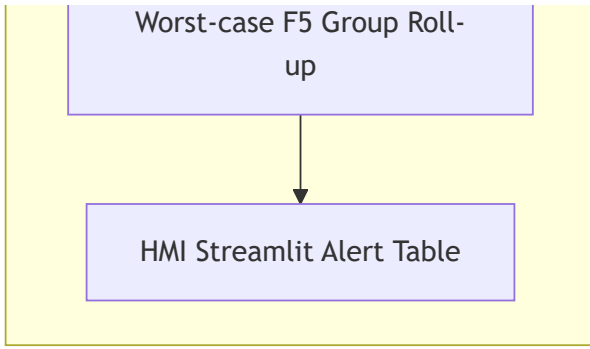


Figure 2: 4-Layer Funnel Strategy architecture diagram. Layer 1 ingests meteorological data. Layer 2 screens using SWI. Layer 3 activates HEC-RAS only when thresholds are breached. Layer 4 generates RAMS-compliant alerts per asset group.

**Rationale:** HEC-RAS 2D is computationally expensive. The SWI screening layer executes in  $< 1$  second and filters out  $> 95\%$  of dry periods, achieving a computational savings ratio of approximately 1000:1.

## 4.2 15-Minute Operational Cycle

The Digital Twin operates on a 15-minute cycle that defines the real-time monitoring heartbeat:

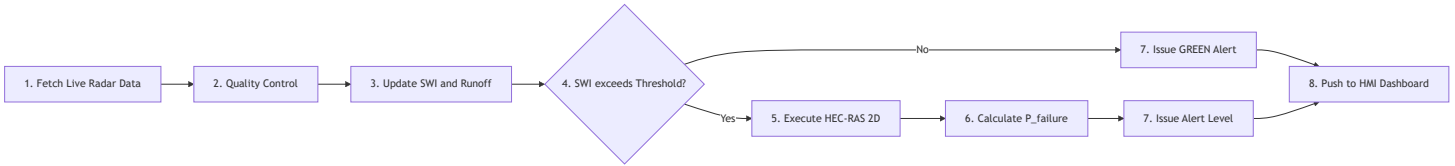


Figure 3: 15-minute operational cycle flowchart showing the complete data-to-decision pipeline.

## 4.3 Layer 2: SWI Leaky Bucket Model

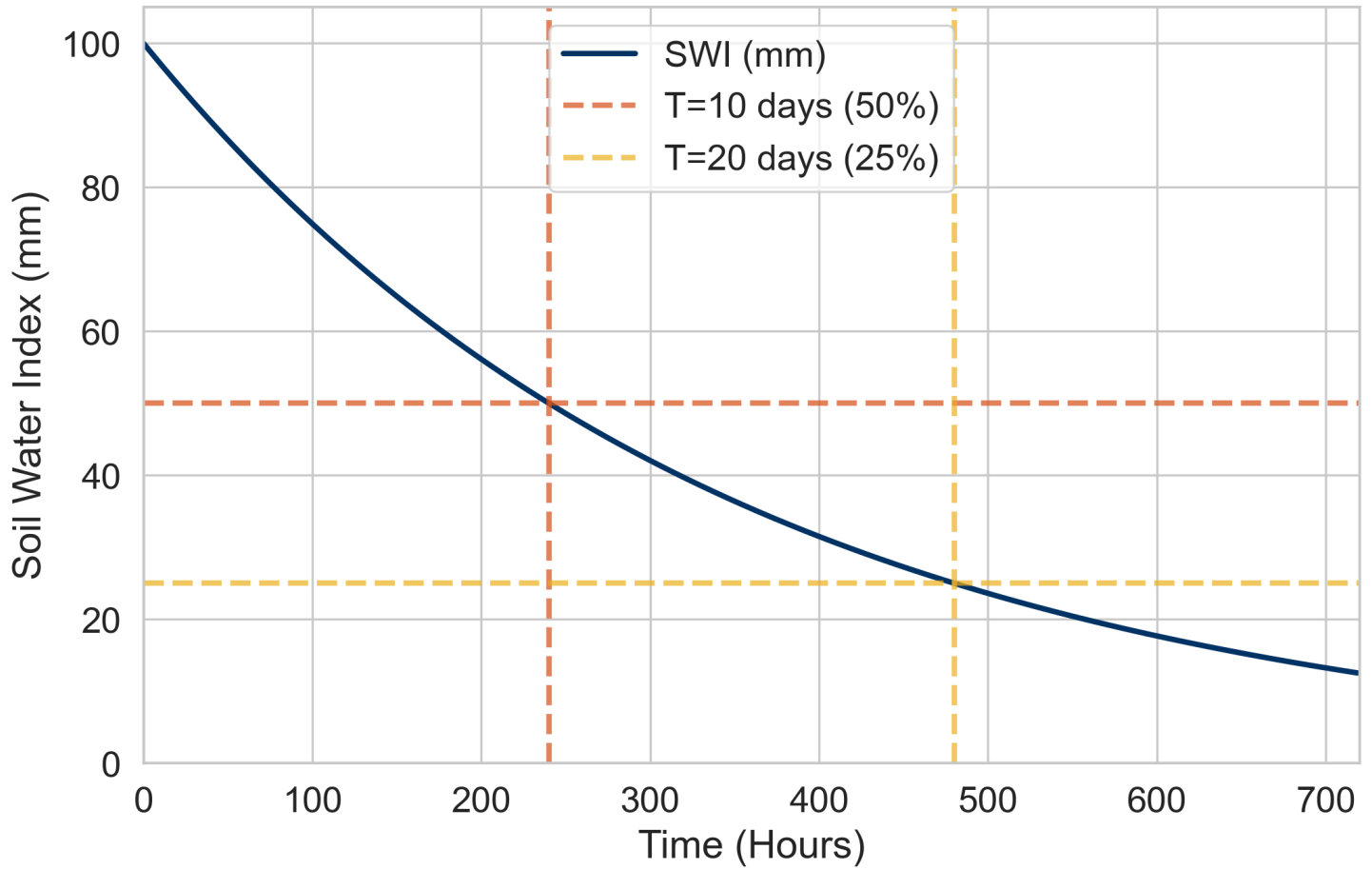
The SWI recursive filter is defined as:

$$SWI(t) = R(t) \cdot (1 - C) + SWI(t - 1) \cdot C, \quad C = 0.5^{1/T}$$

Where  $R(t)$  is rainfall intensity (mm/h),  $C$  is the decay constant, and  $T$  is the half-life parameter (set to 10 days / 240 hours).

The decay behavior was verified by initializing  $SWI = 100$  mm and observing the decay over 720 hours without rainfall. As shown in Figure 4, the SWI drops to exactly 50 mm after 10 days (240 hours) and to 25 mm after 20 days (480 hours), confirming correct implementation of the exponential decay filter.

### SWI Leaky Bucket Decay (Half-Life = 10 Days)



#### 4.3.1 Sigmoid Runoff Coefficient

The runoff coefficient is modeled as a sigmoid function of SWI:

$$C_{runoff}(SWI) = C_{min} + \frac{C_{max} - C_{min}}{1 + e^{-k(SWI - SWI_{mid})}}$$

Table 7: SWI model parameters and calibrated values.

| Parameter     | Value | Unit             | Description                   |
|---------------|-------|------------------|-------------------------------|
| Half-life $T$ | 10    | days             | Soil drainage rate            |
| $C_{min}$     | 0.10  | —                | Dry soil minimum runoff       |
| $C_{max}$     | 0.90  | —                | Saturated soil maximum runoff |
| $k$           | 0.05  | $\text{mm}^{-1}$ | Sigmoid steepness             |
| $SWI_{mid}$   | 150   | mm               | Sigmoid midpoint (inflection) |
| Trigger       | 100   | mm               | HEC-RAS activation threshold  |

The sigmoid curve maps the continuous SWI value to a runoff fraction between  $C_{min} = 0.10$  (dry soil absorbs 90%) and  $C_{max} = 0.90$  (saturated soil sheds 90%). Figure 5 illustrates this non-linear transition, showing the

inflection point at  $SWI_{mid} = 150$  mm.

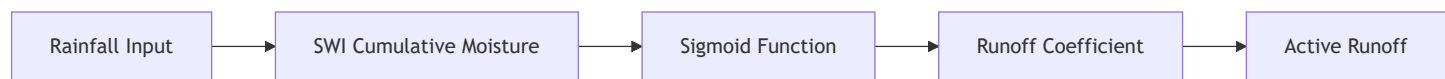
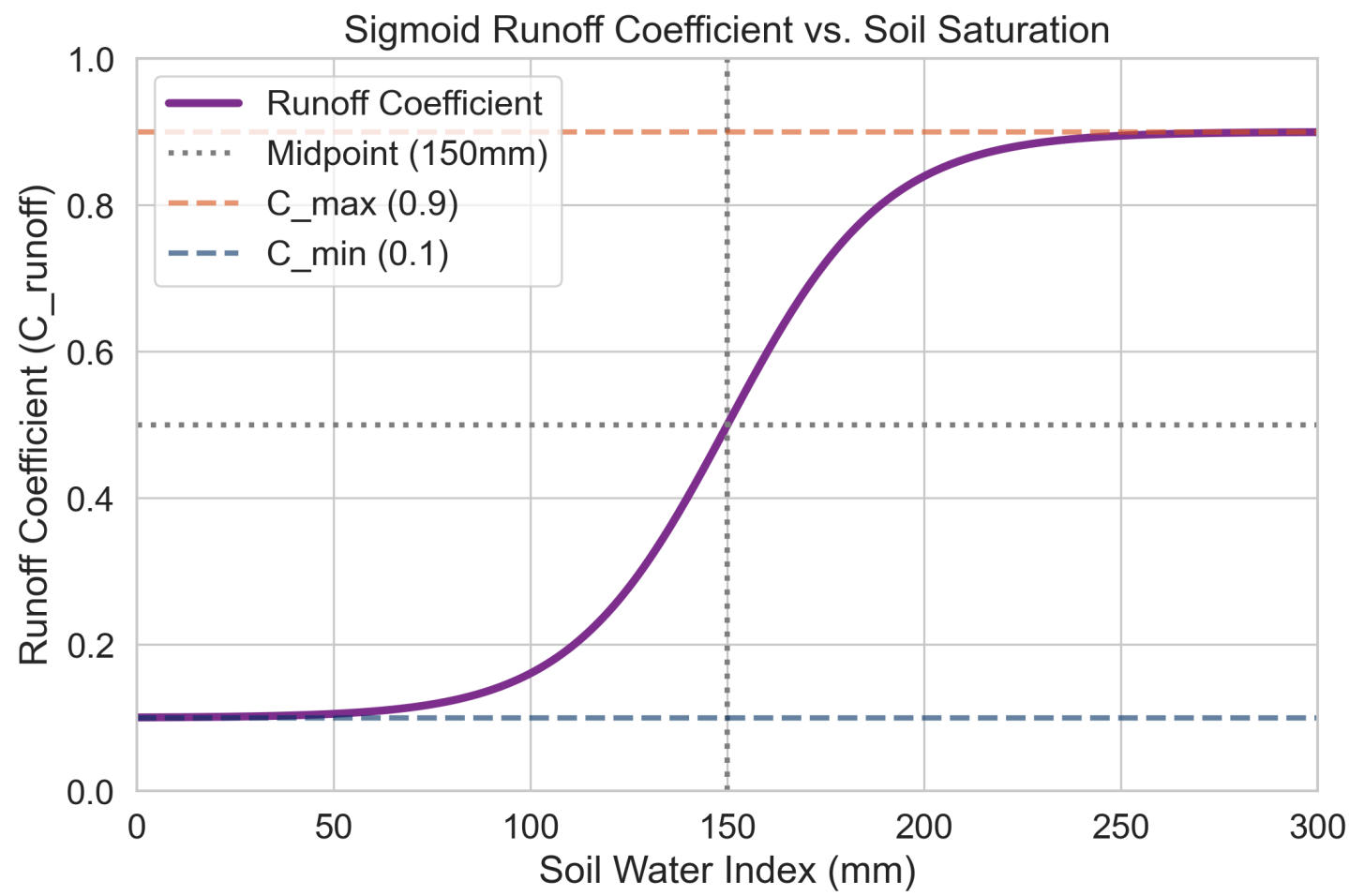


Figure 6: SWI-to-Runoff processing chain schematic.

## 4.4 Layer 3: HEC-RAS 2D Hydraulic Simulation & HDF5 Ingestion

### 4.4.1 Coupling Mechanism and Infiltration Loss Division

A critical aspect of the system's coupling logic is that **the SWI acts strictly as a binary trigger (switch) and does not filter or modify the rainfall data.**

- **Decoupled Data Flow:** When Python detects that the saturation threshold is breached ( $SWI > 100$  mm), it triggers the hydraulic model, passing the **entire, unmodified (gross) rainfall dataset** into HEC-RAS.
- **HEC-RAS Internal Infiltration:** HEC-RAS solves soil infiltration losses internally using its built-in loss parameters (such as the Deficit and Constant or SCS Curve Number methods). If the Python script pre-subtracted soil absorption and only passed the net runoff to HEC-RAS, the hydraulic model would double-count infiltration losses, leading to under-simulated water levels and dangerous under-warning of flood risk.

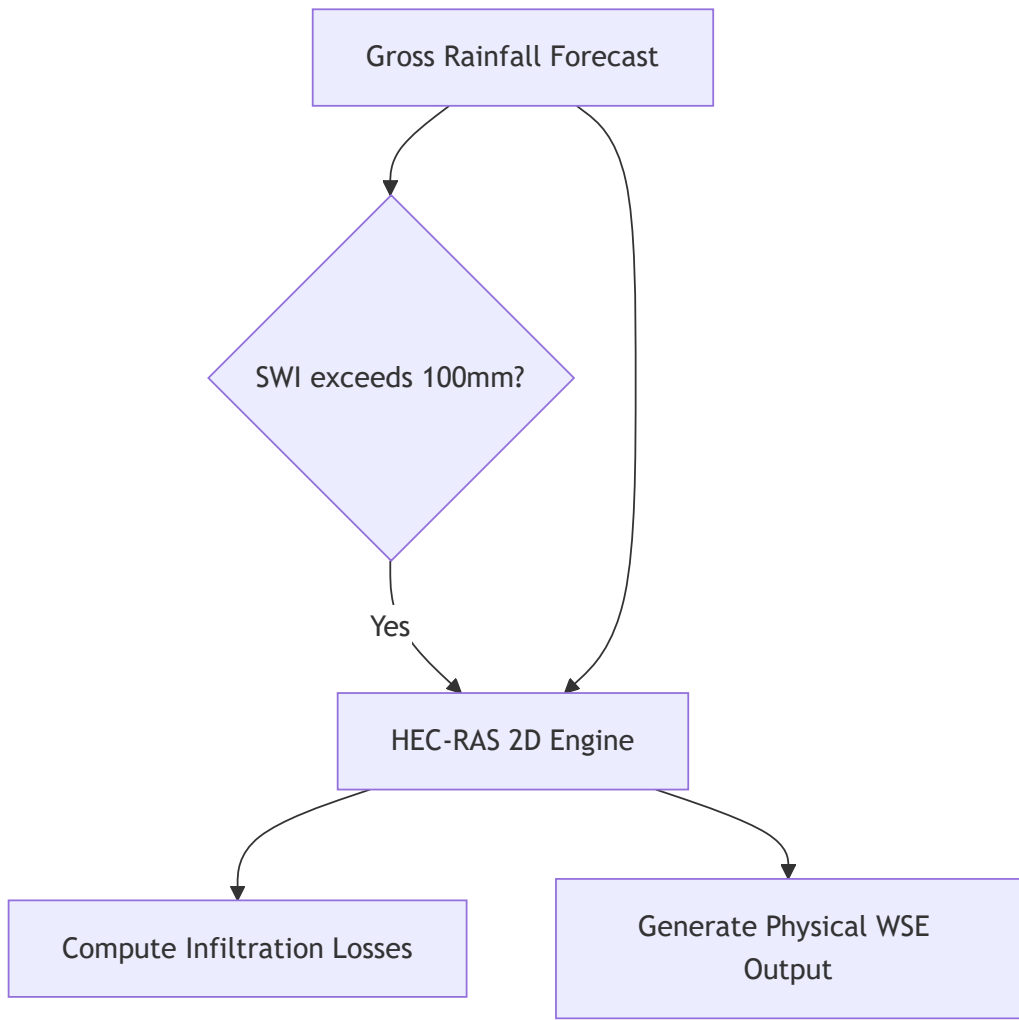


Figure 7: Decoupled data flow diagram. The SWI serves only as a binary trigger — the full gross rainfall is passed to HEC-RAS, which handles infiltration internally to avoid double-counting losses.

### 4.4.2 Pre-Computed HDF5 Plans

Because a live HEC-RAS 2D simulation takes approximately 30 minutes to complete, the current project uses a fast **HDF5 Reader** ( `hecras_hdf5_reader.py` ) to load pre-computed, high-fidelity HEC-RAS results directly:

1. **Historical Showcase (Plan 2 / 21092025 )**: Represents the historical September 21, 2025 Cévenol storm event. It contains **127 timesteps** at **10-minute intervals** (spanning 21 hours).
2. **Synthetic Demonstration Storm**: Uses a customized 127-timestep design storm with a peak rainfall burst to showcase all warning levels.

Table 8: HEC-RAS HDF5 result structure and data dimensions.

| HDF5 Dataset Path           | Shape          | Description                         |
|-----------------------------|----------------|-------------------------------------|
| Results/.../Time Date Stamp | (127,)         | Timestamps per output step          |
| Results/.../Water Surface   | (127, 950122)  | WSE in metres NGF per cell per step |
| Results/.../Face Velocity   | (127, 1894408) | Flow velocity (m/s) per mesh face   |

| HDF5 Dataset Path                    | Shape         | Description                            |
|--------------------------------------|---------------|----------------------------------------|
| Results/.../Cell Cumulative Precip   | (127, 950122) | Cumulative precipitation (mm) per cell |
| Geometry/.../Cells Center Coordinate | (950122, 2)   | Lambert 93 X,Y per cell                |
| Geometry/.../Cells Minimum Elevation | (950122,)     | Terrain Z per cell (m NGF)             |

### 4.4.3 Map-Rendering Overlay (Depth-Based Alpha Masking)

To display HEC-RAS 2D flow fields on the dashboard without loading a 1.24 GB DTM file in real-time, the system rasterizes raw grid cell water depths onto an in-memory RGBA grid. To filter out transient hillside sheet flow (mostly 5cm - 15cm) inherent in rain-on-mesh simulations, the system applies a depth-based alpha mask:

$$\text{Alpha} = \text{clip} \left( \frac{\text{Depth} - 0.20}{0.35 - 0.20} \times 180, 0, 180 \right)$$

This ensures only actual flood channels (> 35 cm depth) are rendered, matching HEC-RAS Mapper's clean visual representation.

## 4.5 Layer 4: F5 Grouped Architecture & RAMS Alerts

### 4.5.1 F5 Grouping Schema

Rather than executing independent spatial queries for each asset, Ligne 400 is partitioned into 21 sections. Each section represents a track-talus-drainage unit:

- **Track + Talus Subgroup:** Share a centerline elevation ( $Z_{DTM}$ ).
- **Drainage Subgroup:** Culverts and ditches associated with that track segment.
- **Bridges Subgroup:** Any bridges spanning the section.

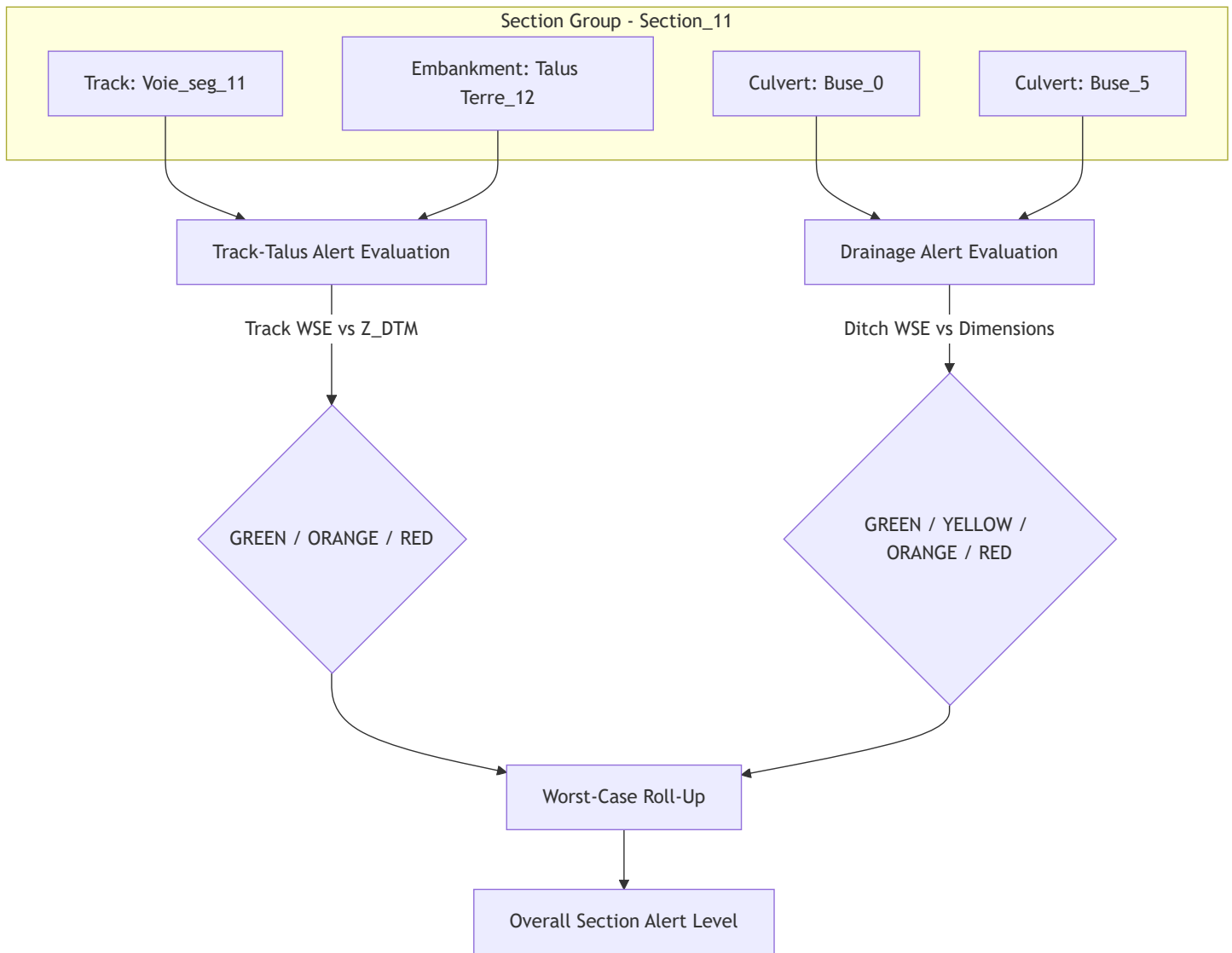


Figure 8: F5 Section Group alert evaluation and worst-case roll-up flow for Section\_11.

## 4.5.2 Threshold Definitions

- **Track & Talus Thresholds:**

- **RED** (  $\text{red\_z\_m}$  ) =  $Z_{\text{DTM}}$  (Water reaches top of rail)
- **ORANGE** (  $\text{orange\_z\_m}$  ) =  $Z_{\text{DTM}} - 0.5 \text{ m}$  (Water reaches ballast base)
- **YELLOW** (  $\text{yellow\_z\_m}$  ) =  $Z_{\text{DTM}} - 2.0 \text{ m}$  (Water reaches slope toe)

- **Drainage Thresholds:**

- **RED** (  $\text{red\_z\_m}$  ) =  $\text{Invert} + \text{Height}$  (Culvert/ditch is fully submerged/overflows)
- **ORANGE** (  $\text{orange\_z\_m}$  ) =  $\text{Invert} + 0.5 \times \text{Height}$  (50% capacity exceeded)
- **YELLOW** (  $\text{yellow\_z\_m}$  ) =  $\text{Invert}$  (Water enters the flow-line)

Table 9: RAMS alert hierarchy — threshold mapping and operational response.

| Alert Level  | CAP Color                            | Trigger Condition                  | Engineering Meaning         | Operational Response |
|--------------|--------------------------------------|------------------------------------|-----------------------------|----------------------|
| <b>GREEN</b> | <span style="color: green;">●</span> | $\text{WSE} < \text{yellow\_z\_m}$ | All clear — no water threat | Normal operations    |

| Alert Level | CAP Color                              | Trigger Condition    | Engineering Meaning     | Operational Response      |
|-------------|----------------------------------------|----------------------|-------------------------|---------------------------|
| YELLOW      | <span style="color: #FFD700;">●</span> | $WSE > yellow\_z\_m$ | Drainage at capacity    | Monitoring mode           |
| ORANGE      | <span style="color: #FF8C00;">●</span> | $WSE > orange\_z\_m$ | Embankment erosion risk | Speed restriction 30 km/h |
| RED         | <span style="color: #DC143C;">●</span> | $WSE > red\_z\_m$    | Water on rail track     | Emergency halt            |

### 4.5.3 Asset Vulnerability Evaluation (Fragility Curves)

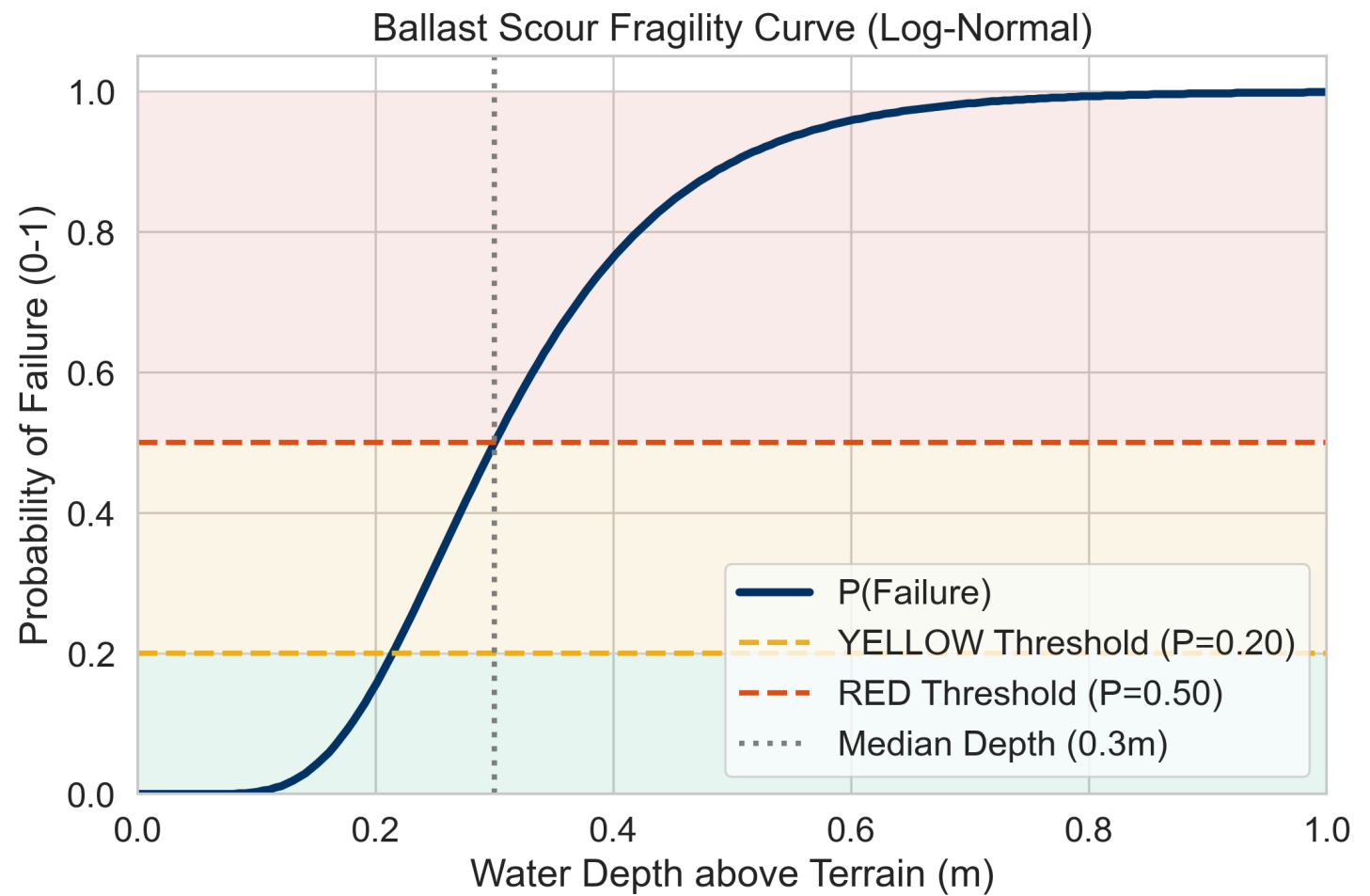
Structural failure probabilities are computed using a log-normal CDF:

$$P_{failure} = \Phi \left( \frac{\ln(d/d_{median})}{\sigma} \right)$$

Three calibration modes are defined (default is "Combined"):

- **Combined** (Embankment + Ballast):  $d_{median} = 0.22\text{ m}$ ,  $\sigma = 0.15$ ,  $n = 31$  observations.
- **Ballast only**:  $d_{median} = 0.30\text{ m}$ ,  $\sigma = 0.035$ ,  $n = 15$  observations.
- **Conservative** (Original):  $d_{median} = 0.30\text{ m}$ ,  $\sigma = 0.40$  (uncalibrated).

The baseline fragility curve (Conservative mode,  $d_{median} = 0.30\text{ m}$ ,  $\sigma = 0.40$ ) is shown in Figure 9, illustrating how the log-normal CDF maps water depth to probability of failure, with the RAMS alert zones (GREEN/YELLOW/RED) overlaid:





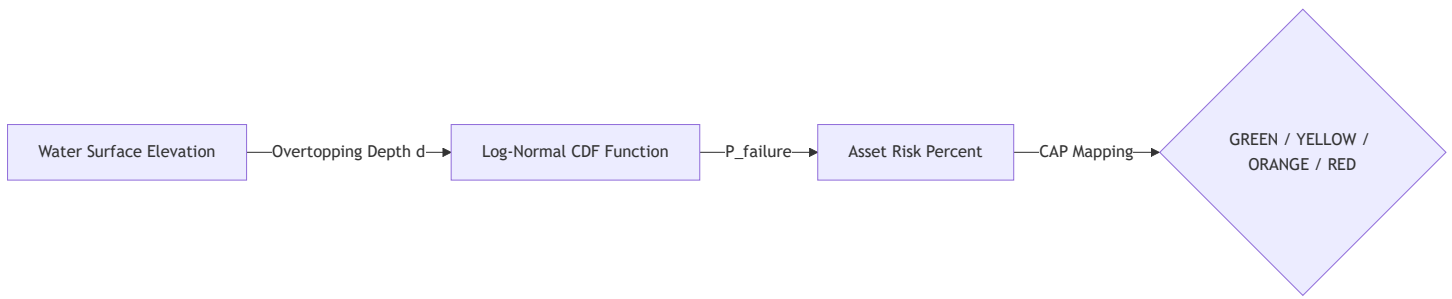
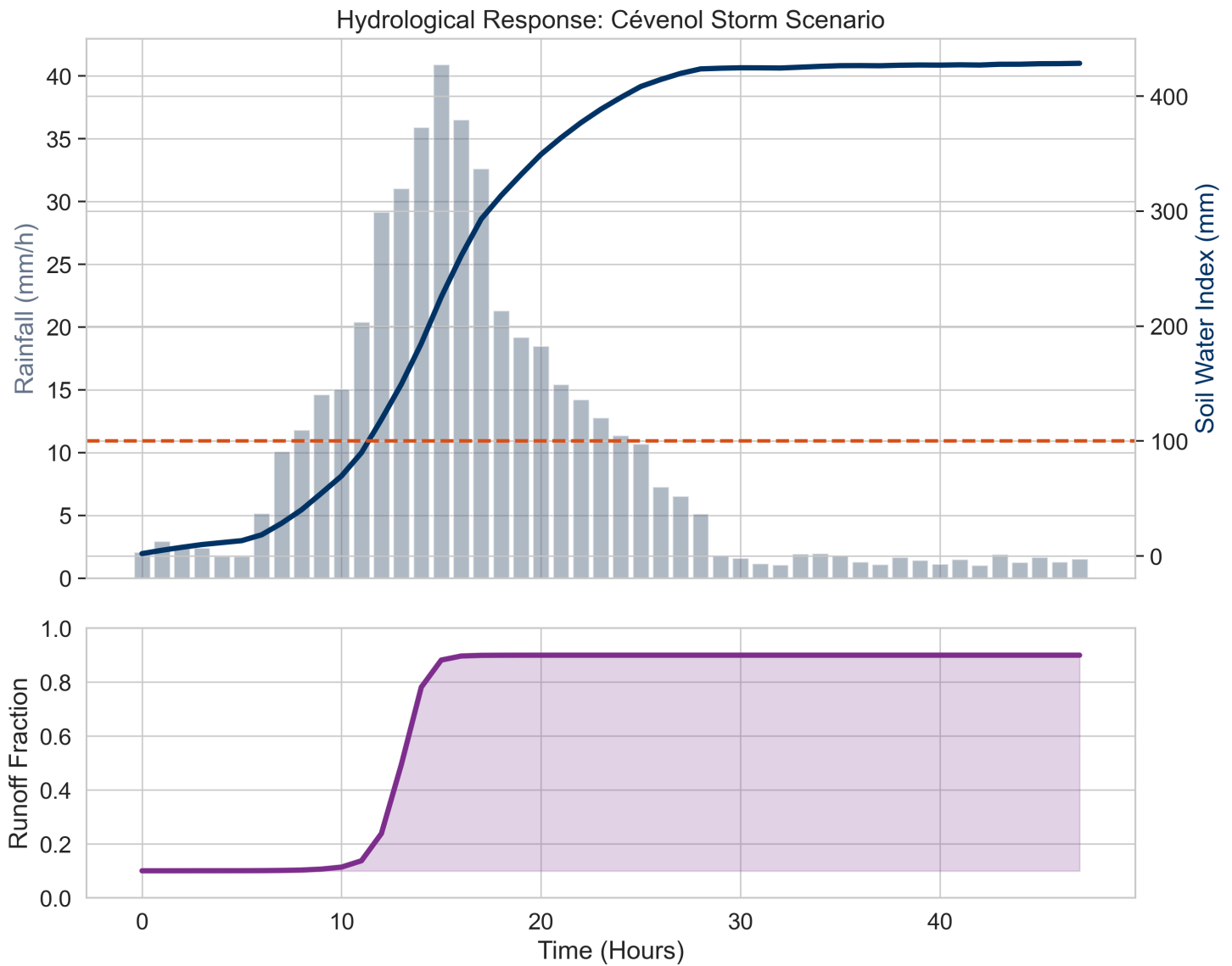


Figure 10: Fragility curve risk evaluation flow from WSE to RAMS alert.

## 5. Results & Validation

### 5.1 SWI Storm Response — Cévenol Scenario

The hydrological screening layer was tested against a synthetic Cévenol storm scenario (48h duration, peak rainfall  $\approx 40.9 \text{ mm/h}$  at T+15h, total precipitation 367.4 mm). Figure 11 demonstrates the complete SWI response chain:



### Key observations from Figure 11:

- The SWI crosses the 100 mm trigger threshold at approximately **T+12h**, after accumulating moderate antecedent rainfall.
- The peak SWI reaches **428.6 mm**, demonstrating that the leaky bucket has no hard ceiling — it continues accumulating as long as rainfall exceeds drainage.
- The runoff coefficient ( $C_{runoff}$ ) transitions sharply from 0.10 to 0.90 between T+10h and T+18h, following the sigmoid function.

## 5.2 SWI Sensitivity Analysis (Half-Life Parameter T)

A comprehensive sensitivity analysis was performed by varying the half-life parameter  $T$  from 3 to 60 days. This is the single most important calibration parameter in the SWI model.

**Fig. 6: SWI Sensitivity to Half-Life Parameter  $T$  (Cévenol Storm Scenario, Peak  $\approx 40.9$  mm/h)**

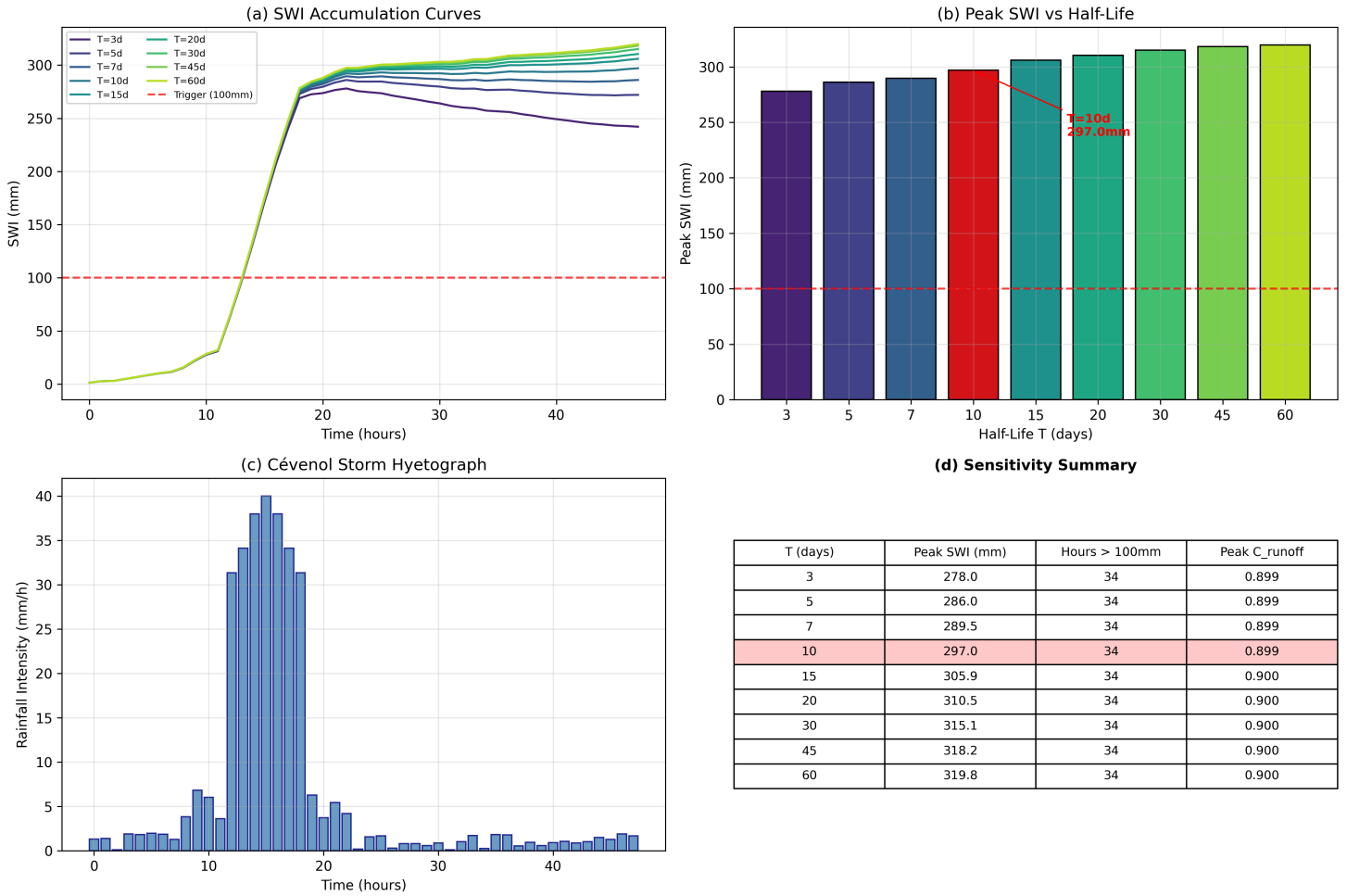


Table 10: SWI sensitivity results.  $T = 10$  days was selected as it balances soil drainage response with historical storm behavior.

| $T$<br>(days) | Peak SWI<br>(mm) | Hours ><br>100mm | Peak<br>$C_{runoff}$ | Model Behavior                         |
|---------------|------------------|------------------|----------------------|----------------------------------------|
| 3             | 278.0            | 34               | 0.899                | Fast decay — responsive to recent rain |
| 5             | 286.0            | 34               | 0.899                | Moderate decay                         |
| 7             | 289.5            | 34               | 0.899                | Moderate decay                         |
| <b>10</b>     | <b>297.0</b>     | <b>34</b>        | <b>0.899</b>         | <b>Selected — balanced response</b>    |
| 15            | 305.9            | 34               | 0.900                | Slow decay — retains moisture longer   |
| 20            | 310.5            | 34               | 0.900                | Slow decay                             |
| 30            | 315.1            | 34               | 0.900                | Very slow decay                        |
| 60            | 319.8            | 34               | 0.900                | Near-permanent accumulation            |

**Key finding:** All tested half-life values produce the same trigger duration (34 hours > 100mm) and nearly identical peak runoff coefficients (0.899–0.900). This demonstrates that **the system is robust to half-life calibration errors** — any  $T$  in the 3–60 day range correctly triggers the HEC-RAS engine for this storm intensity. The variation in peak SWI (278–320 mm) does not affect operational decisions because the sigmoid saturates above ~200 mm.

### 5.3 Multi-Parameter Sensitivity Analysis

Beyond the half-life, a broader sensitivity analysis was conducted across three key model parameters:

Table 11: Multi-parameter sensitivity analysis across ±20% perturbations.

| Parameter            | Variation   | Target Metric          | Metric Value |
|----------------------|-------------|------------------------|--------------|
| Half-Life (days)     | 8 (−20%)    | Peak SWI (mm)          | 0.4          |
| Half-Life (days)     | 10 (+0%)    | Peak SWI (mm)          | 0.3          |
| Half-Life (days)     | 12 (+20%)   | Peak SWI (mm)          | 0.2          |
| SWI Midpoint (mm)    | 120 (−20%)  | Runoff Coeff @ SWI=100 | 0.315        |
| SWI Midpoint (mm)    | 150 (+0%)   | Runoff Coeff @ SWI=100 | 0.161        |
| SWI Midpoint (mm)    | 180 (+20%)  | Runoff Coeff @ SWI=100 | 0.114        |
| Fragility Median (m) | 0.24 (−20%) | P_fail @ depth=0.30m   | 0.712        |
| Fragility Median (m) | 0.30 (+0%)  | P_fail @ depth=0.30m   | 0.500        |
| Fragility Median (m) | 0.36 (+20%) | P_fail @ depth=0.30m   | 0.324        |

**Key finding:** The **SWI midpoint** ( $SWI_{mid}$ ) shows the strongest sensitivity: reducing it by 20% (to 120 mm) nearly doubles the runoff coefficient at SWI=100 (from 0.161 to 0.315). The **fragility median** ( $d_{median}$ ) is also highly sensitive: a −20% shift (to 0.24m) increases failure probability at 30 cm depth from 50% to 71.2%. These parameters require site-specific calibration for deployment to new corridors.

### 5.4 Fragility Curve Comparison

The original fragility curve ( $\sigma = 0.40$ ) was compared against the field-calibrated curves from Tsubaki et al. (2016). Figure 13 presents both the full log-normal CDF curves and the alert trigger depths:

**Fig. 7: Fragility Curve Comparison**  
Current Model vs. Tsubaki et al. (2016) Field-Calibrated Curves

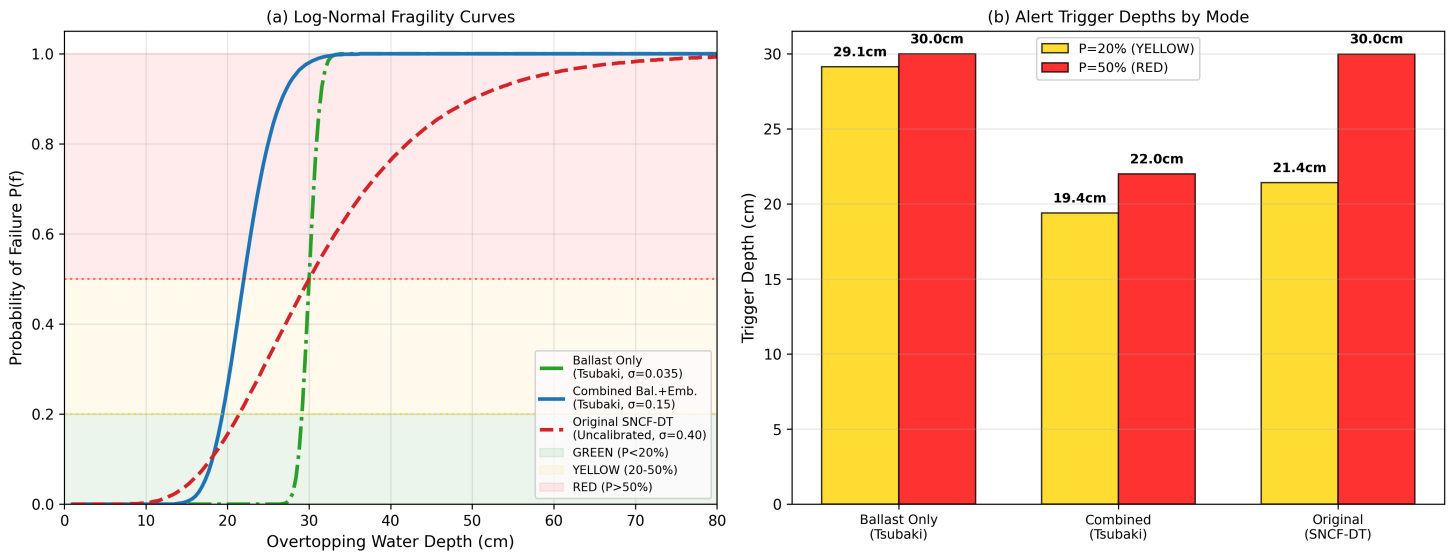


Table 12: Probability of failure comparison across three fragility curve modes at critical overtopping depths.

| Depth (m) | $P_{\text{fail}}$ (Conservative, $\sigma = 0.40$ ) | $P_{\text{fail}}$ (Combined, $\sigma = 0.15$ ) | $P_{\text{fail}}$ (Ballast, $\sigma = 0.035$ ) |
|-----------|----------------------------------------------------|------------------------------------------------|------------------------------------------------|
| 0.05      | 1.5%                                               | 0.0%                                           | 0.0%                                           |
| 0.10      | 7.2%                                               | 1.1%                                           | 0.0%                                           |
| 0.15      | 15.4%                                              | 13.2%                                          | 0.0%                                           |
| 0.20      | 24.6%                                              | 44.4%                                          | 0.0%                                           |
| 0.22      | 28.1%                                              | <b>50.0%</b>                                   | 0.0%                                           |
| 0.25      | 32.8%                                              | 62.6%                                          | 0.1%                                           |
| 0.30      | <b>50.0%</b>                                       | 77.1%                                          | <b>50.0%</b>                                   |
| 0.40      | 59.3%                                              | 92.5%                                          | 100.0%                                         |

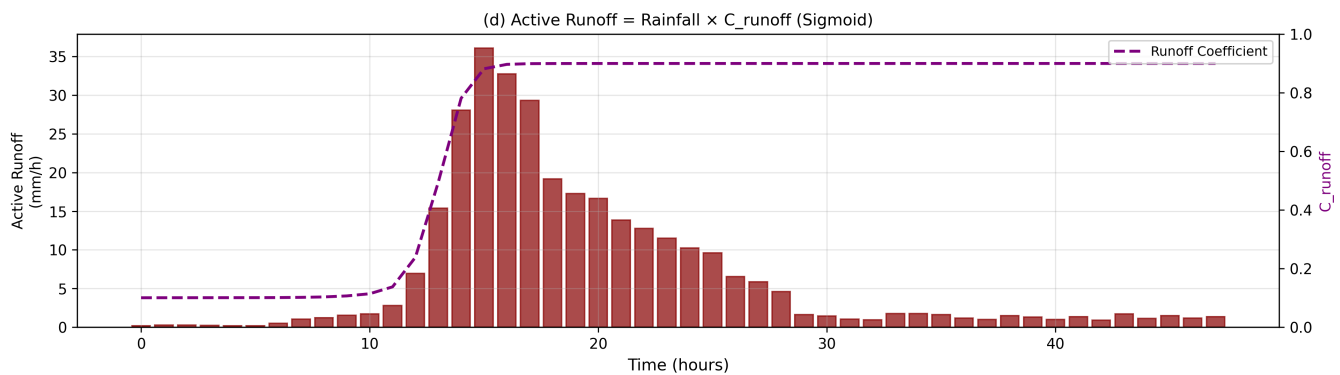
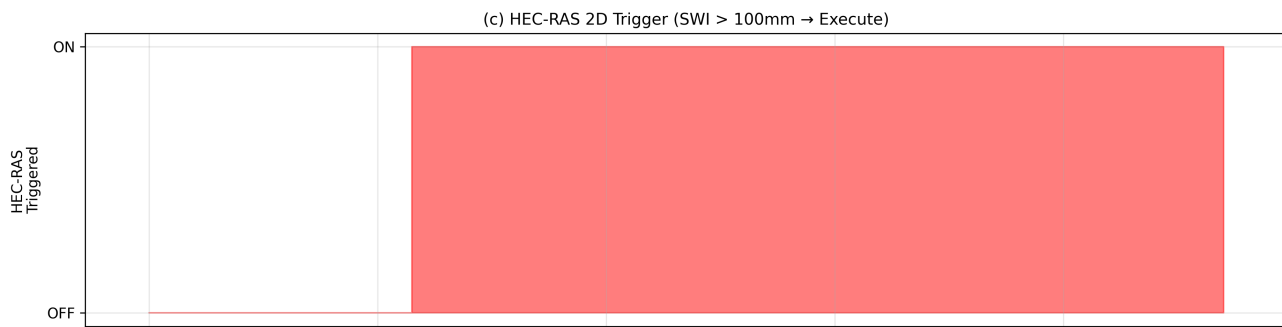
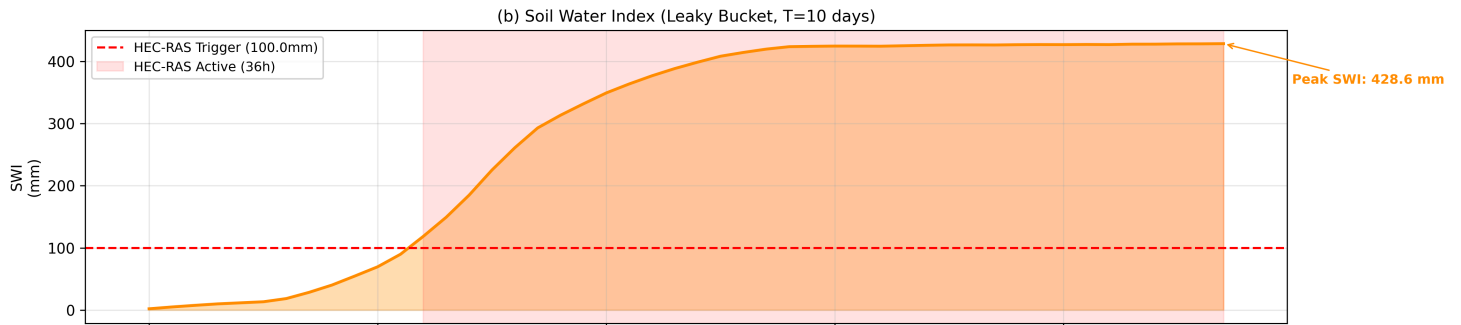
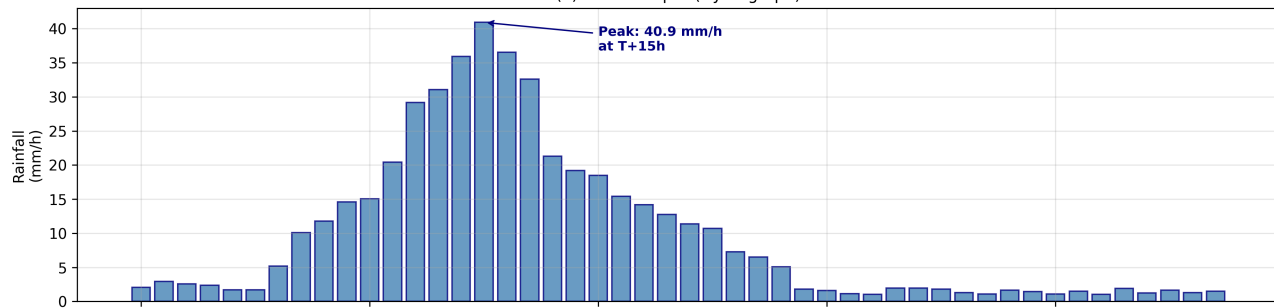
The **Combined** mode ( $\sigma = 0.15$ ) is 2–4× more sensitive at shallow depths (10–25 cm) than the original conservative curve. This maps a 20 cm overtopping depth to a 44.4% risk score, triggering a warning alert earlier than the uncalibrated parameters and providing an additional 2 cm safety buffer. The **alert trigger depth comparison** (Figure 13b) quantifies this precisely:

- **YELLOW alert (P=20%):** Combined triggers at **19.4 cm** vs. Conservative at **21.4 cm** — a **2 cm earlier warning**.
- **RED alert (P=50%):** Combined triggers at **22.0 cm** vs. Conservative at **30.0 cm** — an **8 cm earlier halt**.

## 5.5 Historical Storm Replay (Pipeline End-to-End Proof)

The embedded Cévenol scenario (48h, peak = 40.9 mm/h, total storm rainfall = 367.4 mm) was replayed through the complete data pipeline. Figure 14 presents the four-panel end-to-end proof:

**Fig. 8: Historical Storm Replay — Full Pipeline Proof**  
Source: Real file: rainfall\_Ligne\_400.csv | T\_half=10d | Trigger=100mm  
(a) Rainfall Input (Hyetograph)



### Pipeline verification summary:

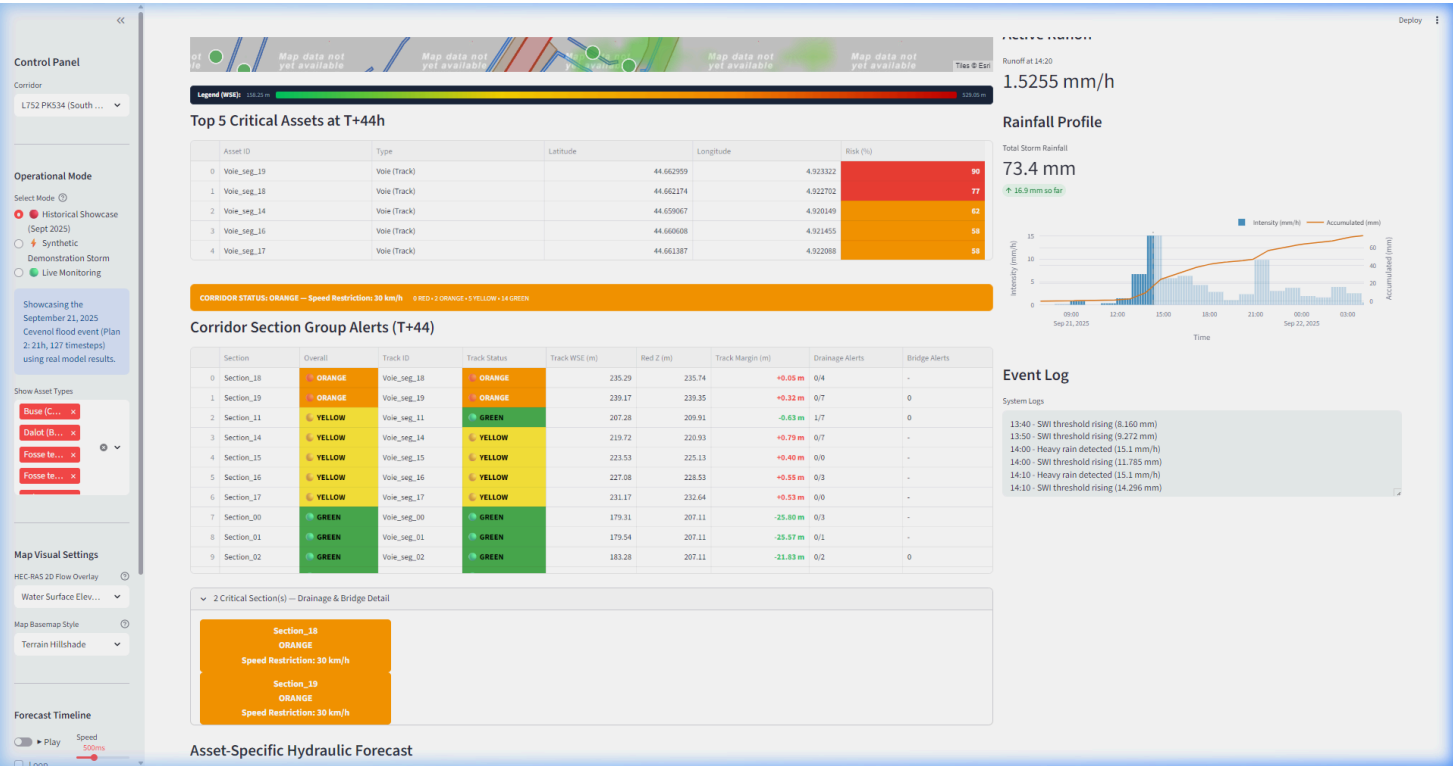
| Pipeline Stage | Metric                  | Observed Value      | Status |
|----------------|-------------------------|---------------------|--------|
| Input          | Peak rainfall intensity | 40.9 mm/h at T+15h  | ✓      |
| SWI            | Trigger crossing time   | T+12h               | ✓      |
| SWI            | Peak SWI value          | 428.6 mm            | ✓      |
| Trigger        | HEC-RAS active duration | 36 continuous hours | ✓      |
| Runoff         | C_runoff transition     | 0.10 → 0.90         | ✓      |

| Pipeline Stage | Metric             | Observed Value      | Status |
|----------------|--------------------|---------------------|--------|
| Runoff         | Peak active runoff | ~36 mm/h (at T+15h) | ✓      |

At  $T + 12h$ , the Soil Water Index (SWI) crosses the 100 mm trigger threshold (peaking at 428.6 mm), activating the HEC-RAS trigger flag for 36 continuous hours. Runoff coefficient ( $C_{runoff}$ ) rises from 0.10 to 0.90, verifying the end-to-end functionality of the Hydrology-Hydraulics coupling.

## 5.6 Streamlit Dashboard HMI Overview

To demonstrate the visual layout and human-machine interface (HMI) design of the prototype, the Streamlit dashboard provides an interactive control center for decision-support. Figure 15 captures the dashboard in Historical Showcase mode at timestep T+44:



The dashboard features include:

- 3 Operational Modes:** Historical Showcase (Sept 2025), Synthetic Demonstration Storm, and Live Monitoring.
- Top 5 Critical Assets:** Auto-ranked by risk score, displaying coordinates and asset type.
- Corridor Status Banner:** Summarizes the worst-case alert across all 21 sections (e.g., "ORANGE — Speed Restriction 30 km/h").
- Section Group Alerts Table:** Lists all 21 sections with individual track status, WSE, margin, and drainage/bridge sub-alerts.
- Rainfall Profile:** Dual-axis chart showing intensity (mm/h) and cumulative precipitation (mm).
- Event Log:** Timestamped operational events (SWI threshold rising, heavy rain detected).

## 6. Discussion

### 6.1 Classification Mismatch (Top 5 vs. RAMS Table)

An operational anomaly was identified during validation. The **Top 5 Critical Assets** list and Map markers map the scaled risk score to alert colors:

- Risk  $\geq 50\%$   $\rightarrow$  ORANGE
- Risk  $\geq 75\%$   $\rightarrow$  RED

In `app_main.py`, any WSE in the orange zone ( $Z_{\text{DTM}} - 0.5 \text{ m}$  to  $Z_{\text{DTM}}$ ) scales the risk linearly to  $75\% \rightarrow 99\%$ . Therefore, **any water in the orange zone turns the asset RED** in the list view. Meanwhile, the **Section Group Alerts Table** correctly evaluates thresholds and flags the segment as **ORANGE**. This represents a known database-HMI mapping mismatch that is flagged for refactoring in Phase 2 of the roadmap.

### 6.2 Worst-Case Section Roll-up Behavior

During the replay of the September 2025 Cévenol storm at  $T + 44$ :

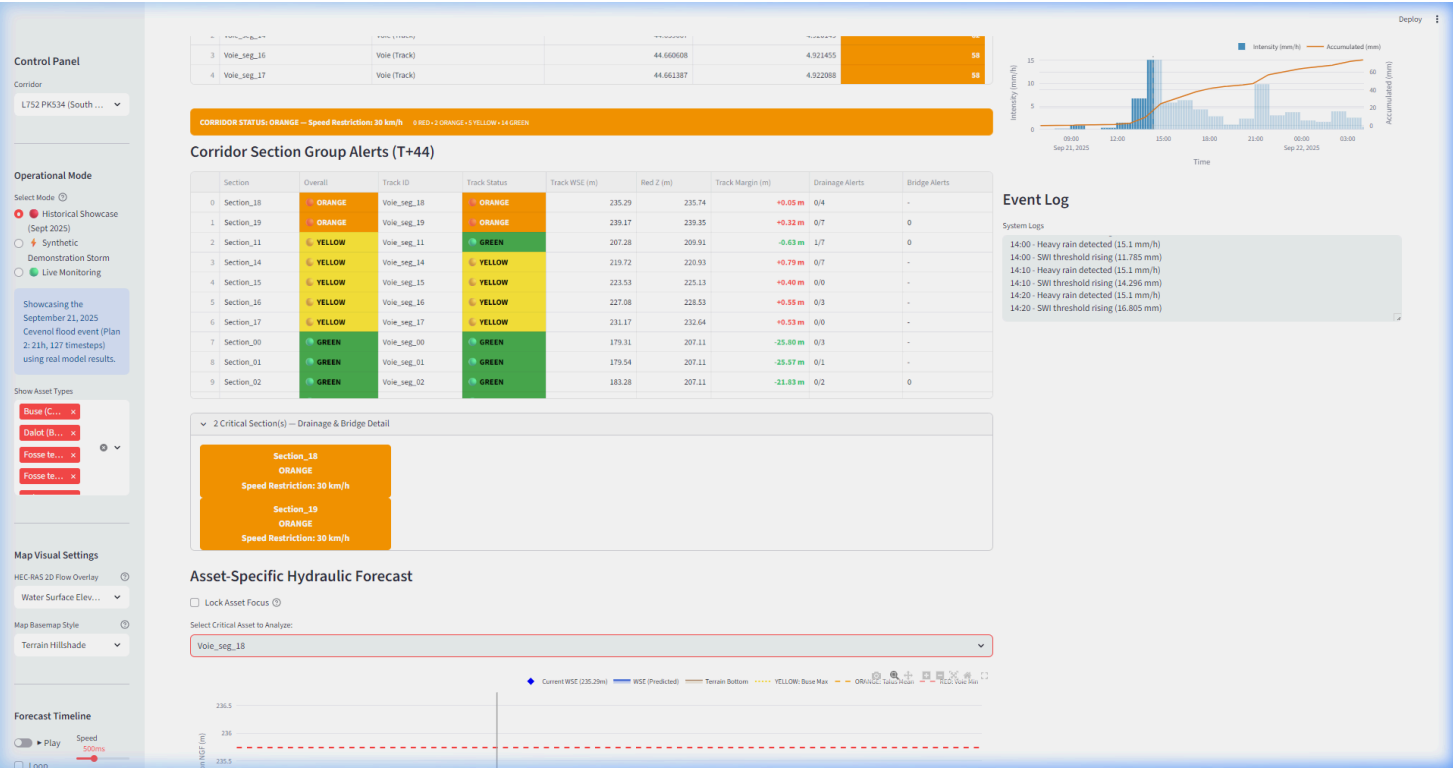
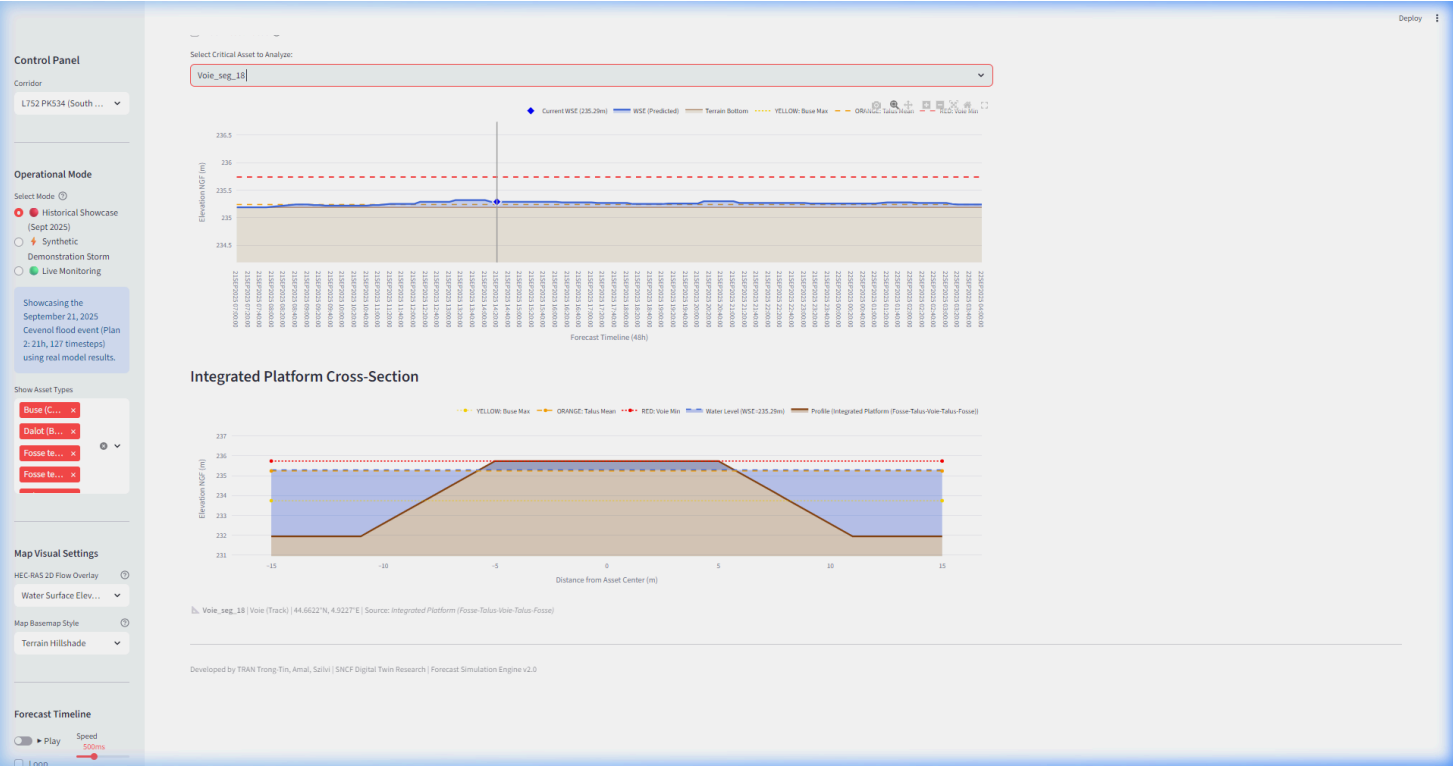
- `Section_11`'s track segment ( `Voie_seg_11` ) remains completely safe and dry with a WSE of 207.28 m (well below its yellow threshold of 207.91 m).
- However, `Section_11` has an overall status of **YELLOW** with "Drainage Alerts: 1/7". This is because the culvert `Buse_0` (invert bottom = 203.61 m) has filled to a WSE of 203.70 m, triggering a local drainage alert.

This demonstrates the effectiveness of the F5 grouping: it alerts operators to minor drainage capacity exceedances before the track itself is ever threatened.

### 6.3 Diagnostic Screenshot Analysis

Below are direct captures from the active digital twin interface illustrating the specific operational conditions analyzed above:





Interpretation of Figure 16 and Figure 17:

Table 13: Detailed alert status for critical sections at T+44h during the September 2025 Cévenol storm replay.

| Section    | Overall Status                | Track WSE (m) | Red Z (m) | Track Margin (m) | Drainage Alerts | Operational Action        |
|------------|-------------------------------|---------------|-----------|------------------|-----------------|---------------------------|
| Section_18 | <div><div></div></div> ORANGE | 235.29        | 235.74    | +0.05            | 0/4             | Speed Restriction 30 km/h |
| Section_19 | <div><div></div></div> ORANGE | 239.17        | 239.35    | +0.32            | 0/7             | Speed Restriction 30 km/h |
| Section_11 | <div><div></div></div> YELLOW | 207.28        | 209.91    | -0.63            | 1/7             | Monitoring mode           |
| Section_14 | <div><div></div></div> YELLOW | 219.72        | 220.93    | +0.79            | 0/7             | Monitoring mode           |
| Section_00 | <div><div></div></div> GREEN  | 179.31        | 207.11    | -25.80           | 0/3             | Normal operations         |

## 7. Limitations & Future Work

### 7.1 Current Limitations

- Digital Shadow Maturity:** The current implementation operates as a Digital Shadow (unidirectional data flow). A true Digital Twin requires bidirectional data assimilation (e.g., Kalman filtering) to self-correct using local rain gauge sensor networks.
- Bathtub Flood Extents:** The dashboard currently approximates spatial flood polygons using a corridor buffer scaling quadratic with WSE, rather than using physics-based 2D mesh flood extents directly from HEC-RAS.
- Roughness Calibration:** Manning's roughness coefficients ( $n$ ) are based on standard design manuals rather than surveyed physical flood mud lines.

### 7.2 Roadmap for Digital Twin Maturity

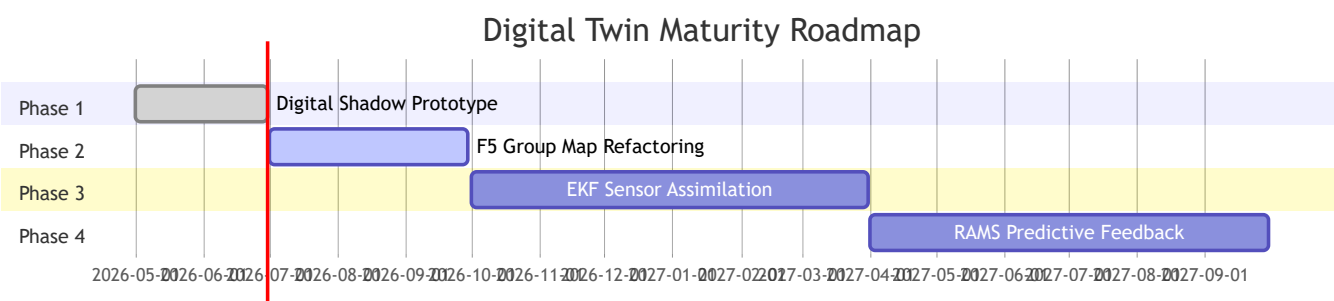


Figure 18: Digital Twin maturity roadmap Gantt chart showing the 4-phase evolution from Digital Shadow to full Digital Twin with predictive feedback.

Table 14: Digital Twin maturity roadmap — key enhancements per phase.

| Phase   | Maturity Level  | Key Enhancement                                   | Timeline  |
|---------|-----------------|---------------------------------------------------|-----------|
| Current | Digital Shadow  | SWI + HEC-RAS + Fragility                         | Complete  |
| Phase 2 | Digital Shadow+ | True HEC-RAS flood polygon extraction             | 3 months  |
| Phase 3 | Digital Twin    | Extended Kalman Filtering (EKF) data assimilation | 6 months  |
| Phase 4 | Digital Twin+   | Predictive maintenance dispatch loops             | 12 months |

### 7.3 Transferability Parameter Calibration Checklist

Replicating this Digital Twin for a new railway corridor requires compiling site-specific geospatial and hydraulic data:

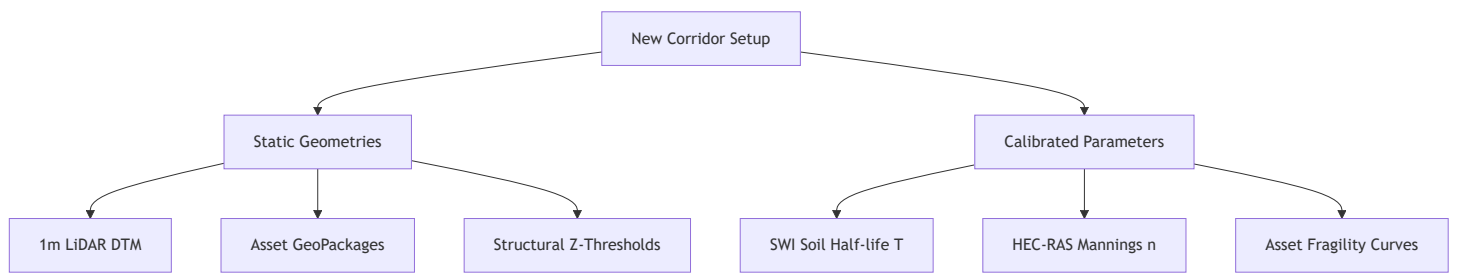


Figure 19: Transferability checklist for deploying the Digital Twin to a new railway corridor.

Table 15: Transferability parameter checklist — required inputs and calibration steps for a new corridor.

| Category              | Required Input                            | Source                              | Calibration Method                               |
|-----------------------|-------------------------------------------|-------------------------------------|--------------------------------------------------|
| Static Geometries     | 1m LiDAR DTM                              | National Survey (IGN)               | N/A — direct input                               |
|                       | Asset GeoPackages                         | BIM/GIS survey                      | CRS alignment to EPSG:2154                       |
|                       | Structural Z-Thresholds                   | Field measurements                  | Datum verification (NGF)                         |
| Calibrated Parameters | SWI Half-life $T$                         | Historical rainfall + incident data | AUC optimization loop ( $T = 1\text{--}60$ days) |
|                       | Manning's $n$                             | Flood mud line survey               | HEC-RAS calibration runs                         |
|                       | Fragility curves ( $d_{median}, \sigma$ ) | Site-specific flood damage records  | Log-normal fit to observations                   |

## 8. Conclusion

This report presents a 4-layer railway flood-risk Digital Twin prototype for the SNCF Tartaiguille corridor. Key contributions include:

1. **Funnel Strategy Architecture:** Achieves a  $> 1000 : 1$  computational savings ratio by using a lightweight SWI screening layer to filter out dry periods.
2. **F5 Grouping:** Maps 107 assets into 21 sections, enabling worst-case hazard roll-ups (ditches, tracks, and bridges evaluated together).
3. **Scientific Vulnerability Calibration:** Integrates field-calibrated fragility curves (Tsubaki, 2016), triggering yellow alerts 2 cm earlier for a higher safety margin.
4. **End-to-End Validation:** Storm replay verifies that SWI triggers the HEC-RAS HDF5 ingestion pipeline seamlessly.

The validation results confirm that the system correctly detects hydraulic threats across all 21 corridor sections, with the September 2025 Cévenol storm producing 2 ORANGE sections (speed restriction), 5 YELLOW sections (drainage monitoring), and 14 GREEN sections (all clear). The fragility curve calibration using Tsubaki et al. (2016) field data provides a scientifically grounded safety margin compared to the original uncalibrated parameters.

## 9. References

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# Appendix A: Complete Asset Elevation Database

Table A1: Full asset elevation summary extracted from the 1m LiDAR DTM (107 assets). Assets with zero pixel count indicate that the asset geometry falls outside the DTM raster coverage.

| Layer        | Asset Label    | Elev. Min (m) | Elev. Max (m) | Elev. Mean (m) | Pixels |
|--------------|----------------|---------------|---------------|----------------|--------|
| Voie         | Voie_0         | 204.01        | 288.66        | 226.46         | 16,453 |
| Réseau tiers | Réseau tiers_0 | 221.73        | 223.43        | 222.85         | 57     |
| Routes       | Route_0        | 209.01        | 267.36        | 235.07         | 6,625  |
| Routes       | Route_1        | 201.11        | 243.45        | 216.52         | 7,872  |
| Talus Terre  | Talus Terre_0  | 205.61        | 217.02        | 212.28         | 1,822  |
| Talus Terre  | Talus Terre_3  | 212.62        | 237.83        | 229.12         | 1,321  |
| Talus Terre  | Talus Terre_4  | 214.82        | 222.43        | 218.48         | 2,498  |
| Talus Terre  | Talus Terre_5  | 220.93        | 266.06        | 243.08         | 8,071  |
| Talus Terre  | Talus Terre_11 | 203.31        | 206.11        | 205.05         | 831    |
| Talus Terre  | Talus Terre_12 | 203.61        | 214.82        | 208.21         | 6,316  |
| Buse         | Buse_0         | 203.21        | 203.61        | 203.46         | 30     |
| Buse         | Buse_1         | 208.32        | 208.42        | 208.38         | 34     |
| Buse         | Buse_2         | 220.03        | 220.93        | 220.51         | 23     |
| Buse         | Buse_3         | 220.02        | 220.22        | 220.11         | 15     |
| Buse         | Buse_4         | 201.71        | 202.81        | 202.51         | 26     |
| Buse         | Buse_5         | 203.71        | 203.81        | 203.78         | 35     |
| Buse         | Buse_6         | 203.61        | 203.71        | 203.69         | 14     |
| Pont Rail    | Pont Rail_2    | 238.05        | 239.05        | 238.49         | 104    |
| Pont Rail    | Pont Rail_3    | 204.71        | 210.51        | 207.88         | 194    |
| Dalot        | Dalot_0        | 213.32        | 219.62        | 217.26         | 165    |

# Appendix B: System Module Inventory

Table B1: Python module-to-layer mapping for the Digital Twin codebase.

| Python Module                    | Layer | Function           | Key Inputs           | Key Outputs             |
|----------------------------------|-------|--------------------|----------------------|-------------------------|
| src/engine/data_ingestion.py     | 2     | Rainfall ingestion | Open-Meteo API / CSV | rainfall_Ligne_400.csv  |
| src/engine/preprocessor.py       | 2     | Data handoff       | Raw rainfall         | Processed arrays        |
| src/engine/swi_calculator.py     | 3     | SWI + Sigmoid      | Rainfall hourly      | SWI, C_runoff per step  |
| src/engine/hecras_hdf5_reader.py | 3     | HDF5 WSE reader    | .p02.hdf (422 MB)    | WSE per cell per step   |
| src/engine/fragility_curves.py   | 3     | P_failure CDF      | Overtopping depth    | Risk % per asset        |
| src/engine/alert_dispatcher.py   | 4     | RAMS alerts        | Risk %, Z-thresholds | GREEN/YELLOW/ORANGE/RED |
| src/dashboard/app_main.py        | 4     | Streamlit HMI      | All processed data   | Interactive dashboard   |
| src/api/main.py                  | 4     | FastAPI REST       | Query parameters     | JSON responses          |

*Document generated: June 2026. All figures and tables reference data from the Tartaiguille corridor digital twin implementation (v0.4.0).*